

Damping Ratio Sets the Response Regime of a Second-Order System:

A Sample Note

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Abstract

This short note is a worked sample of the repository's writing workflow. For the scalar damped harmonic oscillator, the damping ratio ζ alone determines whether the response oscillates. We state the three-regime behaviour and illustrate the underdamped case with a figure generated by the companion experiment `e00001_damped_oscillator`. The note exists to demonstrate structure, not to contribute a new result.

1 Model and Response Regimes

Consider the scalar damped harmonic oscillator

$$\ddot{x}(t) + 2\zeta\omega \dot{x}(t) + \omega^2 x(t) = 0, \quad x(0) = x_0, \quad \dot{x}(0) = 0, \quad (1)$$

with natural angular frequency $\omega > 0$ and damping ratio $\zeta > 0$. The qualitative behaviour of the solution is governed entirely by ζ ; a textbook account is given in [1].

Proposition 1.1 (Response regimes). *Let $x(\cdot)$ solve (1) with $x_0 \neq 0$. Then:*

- (i) *if $\zeta < 1$ (underdamped), x crosses zero infinitely often and decays as $e^{-\zeta\omega t}$, exhibiting overshoot;*
- (ii) *if $\zeta = 1$ (critically damped), x returns to zero without oscillation and with the fastest decay rate among $\zeta \geq 1$;*
- (iii) *if $\zeta > 1$ (overdamped), x returns to zero monotonically but more slowly than the critical case.*

The three cases follow from the roots of the characteristic polynomial $s^2 + 2\zeta\omega s + \omega^2 = 0$, namely $s_{\pm} = \omega(-\zeta \pm \sqrt{\zeta^2 - 1})$, which are complex for $\zeta < 1$, repeated real for $\zeta = 1$, and distinct real for $\zeta > 1$.

Figure 1 shows the underdamped case $\zeta = 0.2$, produced by campaign AA of the companion experiment. Consistent with Proposition 1.1(i), the response overshoots to about -0.53 and settles within $\pm 5\%$ of x_0 after roughly 13.8 time units.

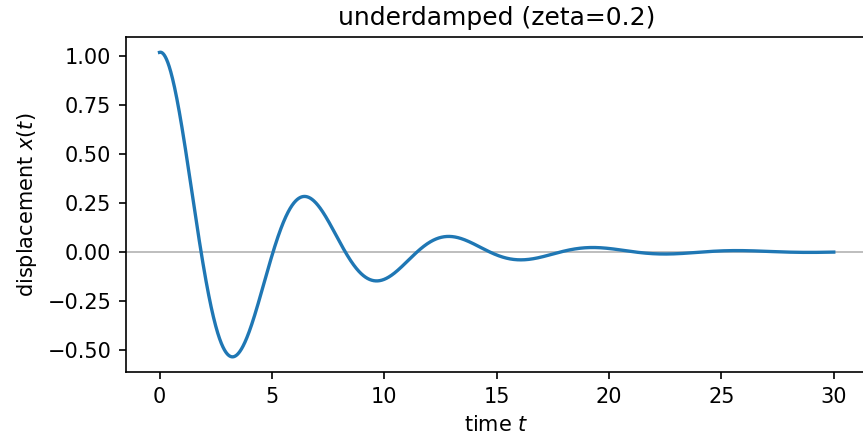


Figure 1: Underdamped response ($\zeta = 0.2$, $\omega = 1$) from (1), generated by `e00001_damped_oscillator` (campaign AA) and copied into this note.

References

- [1] Steven H. Strogatz. *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*. Westview Press, 2nd edition, 2015.